

LARGE ERDŐS–STRAUS WITNESSES II: THE MULTIPLIER MODULUS IS INFINITELY OFTEN $\exp((\log p)^c)$

ANONYMOUS

ABSTRACT. For a prime p let $W(p)$ be the least modulus $M \equiv 3 \pmod{4}$ of a “multiplier” congruence that yields a solution of $4/p = 1/x + 1/y + 1/z$. In earlier work we showed that $W(p)$ exceeds every fixed power of $\log p$ for infinitely many primes p in Mordell’s hard classes. Here we prove $W(p) \geq \exp(c(\log p)^{1/7})$ for infinitely many hard primes, assuming Elsholtz and Tao’s average bound for $\tau(kab^2 + 1)$; more precisely, the least hard prime with $W(p) > T$ is at most $\exp(O((\log T)^7))$. Without the Elsholtz–Tao input we get $\log W(p) \geq (1/\log 2 - o(1)) \log \log p \cdot \log \log \log p$ infinitely often. As before, the class of one modulo the small primes leaves a residual congruence system on the larger primes. The local lemma shows that its avoiding set has Haar density $e^{-O(S)}$, where S is its total mass. To transfer this density to primes we need a *pointwise* minorant that is a short combination of congruence classes, each on few primes. We take the one-sided ℓ^2 sandwich of Bazzi and Razborov from the theory of bounded independence. Its error is the Efron–Stein tail of the avoidance indicator. After a binary encoding of residues the indicator is a CNF of small width, so Håstad’s switching lemma, through the argument of Linial, Mansour and Nisan, bounds the tail. The minorant is transferred to primes by a *linear* transfer theorem. We expand it in Dirichlet characters, bound every coefficient by its Haar ℓ^1 norm, which is within 3% of its mean, and control all characters of conductor up to a power of the modulus at once by Gallagher’s prime number theorem for the family of all characters. The bound for the least such prime is then linear in the logarithm of the largest modulus. No codegree condition, level structure or truncated inclusion–exclusion is needed.

1. INTRODUCTION

The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ is solvable in positive integers for every $n \geq 2$. It suffices to treat primes. It has been verified for $n \leq 10^{18}$ ([24]; [29] for 10^{17}), and Mordell’s identities settle every prime outside the six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$ [23] (cf. [26]). We call the primes in these classes *hard*. For the exceptional set see Vaughan [31] and [27]; for the average number of solutions see [10].

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For $M \equiv 3 \pmod{4}$ put $A_M = (M+1)/4$. If $uvw = A_M$ and $n \equiv -uv^{-1} \pmod{M}$, then $s = (nv+u)/M$ is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsuv} + \frac{1}{nuvw} \quad (1)$$

(direct verification; this is the classical multiplier family, cf. [31, 10]). Define

$$W(n) = \min\{M \equiv 3 \pmod{4} : n \equiv -uv^{-1} \pmod{M} \text{ for some } uvw = A_M\}, \quad (2)$$

with $W(n) = +\infty$ if no such M exists. Thus $W(p) \leq T$ provides an explicit representation with modulus at most T . Every prime $p \equiv 1 \pmod{\text{lcm}\{M \leq T\}}$ has $W(p) > T$ (Lemma 2.2), so Linnik's theorem gives $W(p) \gg \log p$ infinitely often. In [2] we proved that for every fixed k , $W(p) \geq (\log p)^k e^{-C_k \log \log p / \log \log \log p}$ for infinitely many hard primes. Here we prove a sub-exponential rate.

Theorem 1.1 (Proved modulo Theorems 3.2, 2.3 and 6.1). *There is an absolute constant $c > 0$ such that for infinitely many hard primes p*

$$W(p) \geq \exp(c(\log p)^{1/7}).$$

More precisely, for every sufficiently large T there is a prime $p \equiv 1 \pmod{840}$ with $p > T$, $W(p) > T$ and $\log p \leq C(\log T)^7$.

Theorem 1.2 (Proved modulo Theorems 3.2 and 6.1). *For infinitely many hard primes p ,*

$$\log W(p) \geq \left(\frac{1}{\log 2} - o(1)\right) \log \log p \cdot \log \log \log p.$$

Theorem 3.2 is Gallagher's prime number theorem for the family of all primitive characters of conductor at most Q_G [12], quoted from [22, Thm. 28.19] together with the standard facts on exceptional zeros (Theorem 3.1). No hypothesis on Siegel zeros is needed. The first version of this note had the exponent $1/14$ and the constant $1/(2 \log 2)$; it used the Thorner–Zaman theorem [30], through the transfer theorem of [2], in place of Theorem 4.1. [10, Prop. 1.4] is used only to bound the total mass of the residual system by $(\log T)^4 \log \log T$ (Lemma 2.4). Without it, the same argument gives Theorem 1.2. All constants are effective, except that we have not checked whether the implied constant in [10, Prop. 1.4] is. We make no numerical claim.

The method. We fix T and impose $n \equiv 1$ modulo the prime powers $\leq T$ of a set Π of primes, consisting of the primes up to $z = (\log T)^2$ and a few further ones. Then $W(n) > T$ whenever n avoids a finite family of congruence *events*, each on at most $k \approx \log T / (2 \log \log T)$ of the remaining (*free*) primes (§2). In the Haar measure on residues the local lemma shows that the avoiding set has density $\delta \geq e^{-2.2S}$, where $S \ll (\log T)^4 \log \log T$ is the total mass, once an iterated quarantine has made the mass at each free prime small. To find a *prime* in the avoiding set we need a function $B = \sum_i c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ with $B \leq \mathbf{1}[\text{avoid}]$ pointwise, mean $\gg \delta$, Haar ℓ^1

norm $\mathbb{E}|B|$ comparable to its mean, and moduli d_i that each involve few primes. The linear transfer theorem (Theorem 4.1) then gives a prime p with $\log p \ll \log Z$, where Z is the largest modulus. In our construction $\log Z \ll (\log T)^7$.

The transfer theorem is Linnik's theorem with weights. The function $B(n)\mathbf{1}[n \equiv 1 \pmod{Q}]$ is expanded in characters modulo QD , $D = \text{lcm } d_i$. The coefficient of $\chi = \chi_Q \chi_D$ is $\mathbb{E}_D[B\bar{\chi}_D]/\varphi(Q)$, so it is bounded by $\mathbb{E}_D|B|/\varphi(Q)$ whatever the number of cells. Every character that occurs has conductor at most Z , and Gallagher's theorem bounds the sum of $|\vartheta(x; \chi)|$ over *all* of them at once. The exceptional zero is handled as in [2]: either it multiplies the main term, or it belongs to a real character whose coefficient is controlled by a twisted mean of B . For our minorants $\mathbb{E}|B| \leq 1.03 \mathbb{E}B$, because $B \leq F$ and $\mathbb{E}[F - B]$ is small (Lemma 5.5).

In [2] the minorant was an alternating expansion (inclusion–exclusion truncated by the number of primes in the support), organised in levels. That route loses a factorial in the number of levels, which limits it to fixed powers of $\log p$. Here we use instead the one-sided ℓ^2 -approximation scheme of Bazzi [3], in the form of Razborov [28] (Lemma 5.3). For events $E_1, \dots, E_m$ with indicators A_i and arbitrary functions u_j ,

$$B = 1 - \sum_i A_i \left(1 - \sum_{j < i} A_j u_j\right)^2 \leq \mathbf{1}[\text{no } E_i \text{ occurs}],$$

and $\mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$, where $F^{(j)}$ is the indicator that no earlier event occurs, conditioned on E_j . Choosing u_j as the best ℓ^2 approximation of $F^{(j)}$ by sums of functions of at most t coordinates makes every cell of B involve at most $3k + 2t$ primes, and makes the error an *Efron–Stein tail* of $F^{(j)}$. We bound this tail in §6. Encode each residue in binary. Then $1 - F^{(j)}$ becomes a DNF of width $k \log_2(4T^2)$, and Håstad's switching lemma with the Linial–Mansour–Nisan argument [15, 20] gives tail $\leq 4 \cdot 2^{-k_0}$ at level $t \asymp k \log T \cdot k_0$ (Lemma 6.2). We need $k_0 \asymp S + k \log T$, so $t \ll (\log T)^6$. Each modulus then has $\log d_i \ll t \log T \ll (\log T)^7$, and so $\log p \ll (\log T)^7$.

Relation to the literature. The sandwich (Lemma 5.3) is Razborov's identity. The tail bound (Lemma 6.2) is a routine application of [15, 20] (we follow [25, §4]). We claim no novelty for either. To our knowledge these tools from the theory of bounded independence (“polylogarithmic independence fools DNF and AC^0 ” [3, 28, 6]) have not been used before to construct sieve minorants for primes. What makes them usable is the transfer. Gallagher's theorem is classically the engine of Linnik's theorem on the least prime in a single progression. Here it serves as a transfer device for an arbitrary signed combination of progressions. Because it controls the whole family of characters of conductor $\leq Q_G$ simultaneously, the cost is governed by the Haar ℓ^1 norm of the minorant and the size of its largest modulus, and not by the number of cells or their ℓ^1 mass. For a Bazzi–Razborov sandwich the ℓ^1

norm is automatically close to the mean (Lemma 5.5). We do not know an earlier use of Gallagher’s theorem in this role, but we have not searched for one systematically. Linnik-type bounds for primes in sets defined by many congruence conditions are of course classical. Earlier uses of the switching lemma in number theory known to us run in the opposite direction: they show that an arithmetic function such as the Möbius function is uncorrelated with bounded-depth circuits [13, 5]. Our literature search was limited, and we make no priority claim. The Haar-side input, a local-lemma lower bound for the density of the uncovered set of a congruence system, has well-known relatives in the work on covering systems. The closest is Filaseta, Ford, Konyagin, Pomerance and Yu [11], who to our knowledge already use the local lemma to bound the uncovered density of a congruence system from below. Others are Hough [16] and Balister, Bollobás, Morris, Sahasrabudhe and Tiba [4]. We do not use those results. On the bounded-independence side, the step from the truncated inclusion–exclusion of [2] to the present sandwich mirrors a step in that theory. Even, Goldreich, Luby, Nisan and Veličković [9] showed by Bonferroni-type truncation that k -wise independence fools combinatorial rectangles, i.e. conjunctions of single-coordinate conditions (our prime-local case). Bazzi’s sandwich is what extends this to DNFs, i.e. to events on several coordinates.

Status conventions. Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Cited* marks external statements; *Classical* marks standard results for which we include a proof or a precise textbook reference. The cited inputs of the main theorems are Gallagher’s theorem (Theorem 3.2, with the exceptional-zero facts of Theorem 3.1), Elsholtz–Tao (Theorem 2.3) and Håstad’s switching lemma (Theorem 6.1, a textbook result). In the labels, “modulo Theorem 3.2” includes Theorem 3.1. Classical results whose proof we include, or which are standard textbook facts (the local lemma, the Efron–Stein decomposition, the median of the binomial law, the divisor bound), are not listed in the labels. Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large $W(p)$ may well have other representations.

Notation. T is large, $\mathcal{L} = \log T$, $\log_2$ always denotes the binary logarithm (iterated logarithms are written out as $\log \log$), τ is the divisor function and $\tau^*(x) = \max_{n \leq x} \tau(n)$.

2. ATOMS AND THE RESIDUAL SYSTEM

For $M \equiv 3 \pmod{4}$ put $\mathcal{R}(M) = \{-4D \bmod M : D \mid A_M^2\}$. Note that $\gcd(A_M, M) = 1$ (as $4A_M = M + 1$), so every element of $\mathcal{R}(M)$ is a unit class.

Lemma 2.1 (Proved; cf. [2]). $\{-uv^{-1} \bmod M : uvw = A_M\} = \mathcal{R}(M)$. Consequently, for every integer n , $W(n) > T$ if and only if $n \bmod M \notin \mathcal{R}(M)$ for every $M \leq T$ with $M \equiv 3 \pmod{4}$.

Proof. Since $4A_M \equiv 1 \pmod{M}$, from $uvw = A_M$ we get $v^{-1} \equiv 4uw$, hence $-uv^{-1} \equiv -4u^2w$ and $D = u^2w$ divides A_M^2 . Conversely let $D \mid A_M^2$. For each prime q with $q^a \parallel A_M$ and $q^d \parallel D$ ($d \leq 2a$) put $u_q = \max(0, d-a)$, $w_q = d-2u_q$, $v_q = a-u_q-w_q$; these are non-negative, so $u = \prod q^{u_q}$, $v = \prod q^{v_q}$, $w = \prod q^{w_q}$ satisfy $uvw = A_M$ and $u^2w = D$. $\square$

Lemma 2.2 (Class of one; proved). $1 \notin \mathcal{R}(M)$ for every $M \equiv 3 \pmod{4}$.

Proof. If $-4D \equiv 1$ then $D \equiv -A_M \pmod{M}$ (multiply by A_M). Since $4A_M^2 \equiv A_M$, the map $D \mapsto A_M^2/D$ preserves this congruence, so we may assume $D \leq A_M$; then $0 < D + A_M \leq 2A_M < M$, while $M \mid D + A_M$. $\square$

An *atom* is a pair (M, D) with $M \leq T$, $M \equiv 3 \pmod{4}$ and $D \mid A_M^2$. For a set Π of primes write $M = m_\Pi r_\Pi$, where m_Π is the largest divisor of M composed of primes in Π . The atom *survives* Π if $m_\Pi \mid 4D + 1$ and $r_\Pi > 1$; it then defines the *event* $n \equiv -4D \pmod{r_\Pi}$, a unit class, of Haar weight $1/\varphi(r_\Pi)$. Put

$$S^* = S^*(T) = \sum_{(M,D)} \max_{\Pi} \frac{\mathbf{1}[(M,D) \text{ survives } \Pi]}{\varphi(r_\Pi)},$$

the sum over all atoms and the maximum over all sets of primes.

Theorem 2.3 (Elsholtz–Tao [10, Prop. 1.4]; cited). For any $A, B > 1$ and any positive integer $\kappa \ll (AB)^{O(1)}$,

$$\sum_{a \leq A} \sum_{b \leq B} \tau(\kappa ab^2 + 1) \ll AB \log(A+B) \log(1+\kappa).$$

We use it only with $\kappa = 4$, so the factor $\log(1+\kappa) = \log 5$ is a constant.

Lemma 2.4 (Mass bound; proved, and (b) modulo [10, Prop. 1.4]). (a) $S^* \leq \exp((\log 2 + o(1))\mathcal{L}/\log \mathcal{L})$. (b) $S^* \ll \mathcal{L}^4 \log \mathcal{L}$.

Proof. Let (M, D) survive Π , put $A = A_M$ and $g = \gcd(M, 4D + 1)$. Then $m_\Pi \mid g$, and $\varphi(M) \leq m_\Pi \varphi(r_\Pi)$, so $1/\varphi(r_\Pi) \leq g/\varphi(M) \leq C g \log \mathcal{L}/M$ for an absolute C . This bound does not depend on Π , so $S^* \leq C \log \mathcal{L} \sum_{(M,D)} g/M$.

The map $D \mapsto D' = A^2/D$ preserves g . Indeed, D is a unit modulo M , and for a prime power $q^a \mid M$ with $q^a \mid 4D + 1$ we have $4D' \equiv 4A^2 \cdot (-4) \equiv -(4A)^2 \equiv -1 \pmod{q^a}$. It exchanges $D < A$ with $D > A$, so it suffices to sum over $D \leq A$ and double. Write $D = sr^2$ with s squarefree. Then $D \mid A^2$ if and only if $sr \mid A$; write $A = srh$, so that $D \leq A$ if and only if $r \leq h$, and $M = 4srh - 1 \geq 3srh$. From $g \mid 4sr^2 + 1$ and $g \mid M = 4srh - 1$ we get $g \mid 4sr(r+h)$, and $\gcd(g, 4sr) = 1$, so $g \mid \gcd(4sr^2 + 1, r+h)$. For fixed

(s, r) the triple (M, D, g) is determined by h . Using $g \leq \sum_{e|g} e$, we get

$$\sum_{r \leq h \leq X} \frac{g}{h} \leq \sum_{e|4sr^2+1} e \sum_{\substack{r \leq h \leq X \\ h \equiv -r \pmod{e}}} \frac{1}{h} \leq (3 + \log X) \tau(4sr^2 + 1), \quad X = \frac{T+1}{4}.$$

For the last step: the least admissible h is at least $\max(r, e - r) \geq e/2$, so the inner sum is at most $2/e + (1 + \log X)/e$. Hence

$$S^* \leq C' \log \mathcal{L} \cdot \mathcal{L} \sum_{sr^2 \leq X} \frac{\tau(4sr^2 + 1)}{sr}.$$

(a) Bound $\tau(4sr^2 + 1) \leq \tau^*(T + 2)$ and $\sum_{s, r \leq X} 1/(sr) \leq (1 + \mathcal{L})^2$. Then use $\log \tau^*(x) \leq (\log 2 + o(1)) \log x / \log \log x$ [17, Thm. 317]. (b) Split $s \in [2^i, 2^{i+1})$, $r \in [2^j, 2^{j+1})$ with $2^i, 2^j \leq X$. By Theorem 2.3 with $\kappa = 4$, $A = 2^{i+1}$, $B = 2^{j+1}$ (so $A, B > 1$ and $\kappa \ll (AB)^{O(1)}$), each block contributes $\ll 2^{-i-j} AB \log(A + B) \log 5 \ll \mathcal{L}$. There are $\ll \mathcal{L}^2$ blocks, so the sum is $\ll \mathcal{L}^3$. $\square$

Lemma 2.5 (Iterated quarantine; proved, cf. [2]). *Let $2 \leq z \leq T$, $k = \lfloor \mathcal{L}/\log z \rfloor$ and $0 < c_0 \leq 1$. For a set $\Pi \supseteq \{\ell \leq z\}$ and a prime $\ell \notin \Pi$ let $w_\ell(\Pi)$ be the total Haar weight of the distinct events of atoms surviving Π whose modulus r_Π is divisible by ℓ . Start from $\Pi_0 = \{\ell \leq z\}$ and repeatedly add to Π every prime $\ell \notin \Pi$ with $w_\ell(\Pi) > c_0$. The process stops at some $\Pi = \Pi_0 \cup \mathcal{B}$ with $|\mathcal{B}| \leq kS^*/c_0$, and then $w_\ell(\Pi) \leq c_0$ for every prime $\ell \notin \Pi$.*

Proof. The process stops since only primes $\leq T$ can have $w_\ell > 0$. If ℓ is added at a stage Π_i , then $c_0 < w_\ell(\Pi_i)$, which is at most the sum of the contributions to S^* of the atoms (M, D) with $\ell \mid M$. Summing over the added primes, each atom is counted at most once for each prime factor $\ell > z$ of M . There are at most k such factors since $M \leq T$. $\square$

Definition 2.6 (Residual system). Let $\Pi \supseteq \{\ell \leq z\}$ be as in Lemma 2.5, let $e_\ell \geq 1$ be maximal with $\ell^{e_\ell} \leq T$, and put $Q_\Pi = \text{lcm}(24, \ell^{e_\ell} : \ell \in \Pi, \ell \leq T)$. The *free* primes are $\mathcal{P} = \{\ell \leq T : \ell \notin \Pi\}$, with *coordinates* $X_\ell = n \bmod \ell^{e_\ell}$ taking values in $G_\ell = (\mathbb{Z}/\ell^{e_\ell})^\times$. The *Haar measure* $\mathbb{P}$ makes the X_ℓ independent and uniform. For each *distinct* pair $(r, c) = (r_\Pi, -4D \bmod r_\Pi)$ arising from an atom (M, D) surviving Π , with $r = \prod \ell^{a_\ell}$, and each *lift*, i.e. each $\sigma \in \prod_{\ell \mid r} G_\ell$ with $\sigma_\ell \equiv c \pmod{\ell^{a_\ell}}$, we take the event $E = \{X_\ell = \sigma_\ell \text{ for all } \ell \mid r_\Pi\}$, with support $\text{supp } E = \{\ell \mid r\}$. Let $E_1, \dots, E_m$ list these events (two pairs may have a common lift; such duplicates are kept), and let $F = \mathbf{1}[\text{no } E_i \text{ occurs}]$.

The lifts of one pair (r, c) partition the class $c \bmod r$, so their weights add up to $1/\varphi(r)$. This pair-then-lift construction is what makes Lemma 2.7(iii) an equality: duplicate lifts of different pairs are counted in w_ℓ and in $w_\ell(\Pi)$ alike. Duplicates are harmless in Lemmas 5.1, 5.3 and 5.6, which are stated for indexed families.

Lemma 2.7 (Properties of the residual system; proved). *In Definition 2.6, with $z \geq 7$:*

- (i) $840 \mid Q_\Pi$ and $\log Q_\Pi \leq (\pi(z) + |\mathcal{B}|)\mathcal{L} + \log 24$;
- (ii) $|\mathcal{P}| \leq T$, $|G_\ell| \leq T$, $|\text{supp } E_i| \leq k$, and $m \leq T^{k+2}$;
- (iii) $w_\ell := \sum_{i: \ell \in \text{supp } E_i} \mathbb{P}(E_i) = w_\ell(\Pi)$ for every $\ell \in \mathcal{P}$, and $S := \sum_{i \leq m} \mathbb{P}(E_i) \leq S^*$; so if Π is the output of Lemma 2.5, then $w_\ell \leq c_0$ for all $\ell \in \mathcal{P}$;
- (iv) (Reduction.) If $n \equiv 1 \pmod{Q_\Pi}$ and $F(n) = 1$ (the residues $X_\ell = n \bmod \ell^{e_\ell}$ need not be units), then $W(n) > T$.

Proof. (i) $24 \mid Q_\Pi$ and $5, 7 \in \Pi$. (ii) An event's support consists of the prime factors $> z$ of some $M \leq T$, so there are at most k of them. There are at most T^2 atoms, since $\tau(A_M^2) \leq 2A_M \leq T$, and each has at most $\prod_{\ell \mid r_\Pi} \ell^{e_\ell - a_\ell} \leq T^k$ lifts. (iii) Lifting preserves weights. Each pair (r, c) comes from an atom whose contribution to S^* is at least $1/\varphi(r)$. (iv) Let $M \leq T$ with $M \equiv 3 \pmod{4}$, and suppose $n \equiv -4D \pmod{M}$ with $D \mid A_M^2$. Every $\ell^v \parallel M$ with $\ell \in \Pi$ has $\ell^v \leq T$, so $m_\Pi \mid Q_\Pi$ and $-4D \equiv n \equiv 1 \pmod{m_\Pi}$, i.e. $m_\Pi \mid 4D + 1$. If $r_\Pi = 1$ this contradicts Lemma 2.2. Otherwise the atom survives Π , and $n \bmod \ell^{e_\ell}$ (a unit, as $-4D$ is) is a lift of $-4D \bmod \ell^{a_\ell}$ for each $\ell \mid r_\Pi$. So a lift of the pair $(r_\Pi, -4D \bmod r_\Pi)$ occurs, contradicting $F(n) = 1$. Conclude with Lemma 2.1. $\square$

3. THE ANALYTIC INPUT: GALLAGHER'S THEOREM

For a Dirichlet character χ put $\vartheta(x; \chi) = \sum_{p \leq x} \chi(p) \log p$, and let $E_0(\chi) = 1$ if χ is the trivial character modulo 1 and $E_0(\chi) = 0$ otherwise. $\sum_{\chi \bmod q}^*$ is a sum over primitive characters.

Theorem 3.1 (Landau–Page; classical, [7, Ch. 14], [21, Cor. 11.10]). *There are absolute effective positive constants c_1, c_2 such that, for every $Y \geq 3$, $\prod_{q \leq Y} \prod_{\chi \bmod q}^* L(s, \chi)$ has at most one zero in the region $\Re s > 1 - c_1/\log Y$, $|\Im s| \leq Y$. If this zero β_1 exists, it is real and simple, and its character χ_1 is quadratic. Moreover, every real zero β of $L(s, \chi)$, for χ a real primitive character of conductor $q \geq 3$, satisfies $1 - \beta \geq c_2^{-1} q^{-1/2} (\log q)^{-2}$.*

The first part is the classical zero-free region with the Landau–Page theorem. The height restriction matters: without it the statement would be a zero-free region of width $\asymp 1/\log Y$ uniformly in $|\Im s|$, which is not known. The “Exceptional Zero Statement” (28.61)–(28.62) of [22], which refers to [21, Cor. 11.10], omits the height restriction. The second part is the effective Page-type bound [7, Ch. 14], [22, (28.62)]. We use Theorem 3.1 only for *real* zeros: for the uniqueness of a real zero close to 1, for the quadraticity of its character, and for the lower bound for $1 - \beta$.

Theorem 3.2 (Gallagher [12]; cited in the form of [22, Thm. 28.19]). *There are absolute constants $c \geq 1$ and $\kappa_0 \geq 3$ with the following property. Let*

$\kappa \geq \kappa_0$ be fixed and $1 < Q_G^{6c} \leq x$. Then

$$\sum_{q \leq Q_G} \sum_{\chi \bmod q}^* |\vartheta(x; \chi) - E_0(\chi)x| \ll_{\kappa} x \exp\left(-\frac{\log x}{\kappa \log Q_G}\right) + \left(\frac{\log x}{\log Q_G}\right)^2 \frac{x}{Q_G},$$

unless $\prod_{q \leq Q_G} \prod_{\chi \bmod q}^* L(s, \chi)$ has a real zero β_1 with $1 - \beta_1 < 1/(\kappa \log Q_G)$. In that case the term of the exceptional character χ_1 on the left is replaced by $|\vartheta(x; \chi_1) + x^{\beta_1}/\beta_1|$, and the right-hand side by

$$(1 - \beta_1)(\log x) \left(x \exp\left(-\frac{\log x}{\log Q_G}\right) + \frac{x \log x}{Q_G \log Q_G} \right).$$

Source caveat. We quote [22], an unpublished draft (third volume of Montgomery–Vaughan), archived by us. We have not seen Gallagher’s paper [12], so we cite it without a theorem number. The printed proof of [22, Thm. 28.19] has visible slips. Its case split is at $1 - \beta_1 < 1/(3\kappa_0 \log Q_G)$, so it establishes the statement for $\kappa = 3\kappa_0$. The statement for every larger κ follows from that case. Let $\kappa \geq 3\kappa_0$, and suppose a real zero satisfies $1/(\kappa \log Q_G) \leq 1 - \beta_1 < 1/(3\kappa_0 \log Q_G)$. Then $x^{\beta_1}/\beta_1 \leq 2x \exp(-\log x/(\kappa \log Q_G))$, and $(1 - \beta_1)(\log x) x \exp(-\log x/\log Q_G) \ll x \exp(-\log x/(2 \log Q_G))$. So the exceptional form of the bound implies the non-exceptional one with parameter κ . We therefore fix

$$\kappa := \max(3\kappa_0, 1/c_1).$$

An exceptional zero as in Theorem 3.2 is then a real zero in the region of Theorem 3.1 with $Y = Q_G$, so it is unique and χ_1 is real. Write C_G for the implied constant in Theorem 3.2 with this κ .

Gallagher’s original statement may carry a range condition of the form $\exp(\sqrt{\log x}) \leq Q_G \leq x^b$; we have not checked this. Our use is robust to such a condition. In the proof of Theorem 4.1, $\log x / \log Q_G = \kappa L \geq 6c$, and $\log x \geq (\kappa L)^2$ by the hypothesis of that theorem, so $Q_G \geq \exp(\sqrt{\log x})$. The statement itself is the standard form of Gallagher’s prime number theorem for the whole family of characters of conductor $\leq Q_G$, which underlies the usual proofs of Linnik’s theorem.

4. A LINEAR TRANSFER THEOREM

For a modulus D write $\mathbb{E}_D$ for the mean over the units modulo D .

Theorem 4.1 (Linear transfer; proved modulo Theorems 3.1 and 3.2). *Let $T \geq 1$ and $Q \geq 2$. Let $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ be a finite real combination with $\gcd(d_i, Q) = \gcd(b_i, d_i) = 1$, and assume that $B(n) \leq \mathbf{1}[W(n) > T]$ for every $n \equiv 1 \pmod{Q}$ coprime to all d_i . Put*

$$D = \text{lcm}_i d_i, \quad Z = Q \max_i d_i, \quad \mu = \mathbb{E}_D B = \sum_i \frac{c_i}{\varphi(d_i)}, \quad A = \frac{\mathbb{E}_D |B|}{\mu},$$

and assume $\mu > 0$, $A \leq Z^{1/4}$, and the twist condition: for every real primitive character ψ of conductor $f > 1$ with $\gcd(f, Q) = 1$ and $f \mid d_i$ for some i ,

$$|\mu_\psi| \leq \mu/4, \quad \mu_\psi = \sum_{i: f \mid d_i} c_i \psi(b_i) / \varphi(d_i).$$

There is an absolute effective constant C_2 such that, if

$$\log x \geq C_2(1 + \log A) \log Z,$$

then some prime $p \leq x$ with $p \equiv 1 \pmod{Q}$ and $p \nmid D$ has $W(p) > T$.

Neither the ℓ^1 mass $\sum_i |c_i| / \varphi(d_i)$ nor the number of cells enters; only the Haar quantity $A \geq 1$ does.

Proof. Parameters. Put $L = 2c + \log(400C_G(A + 1))$ and $Q_G = x^{1/(\kappa L)}$. Since $A \geq 1$ we have $L \leq C_3(1 + \log A)$ for an absolute C_3 . If C_2 is large enough, the hypothesis gives the following, using $A \leq Z^{1/4}$ and $Z \geq 2$: $\log Q_G \geq (C_2/(\kappa C_3)) \log Z$, hence $Q_G \geq Z$ and $Q_G \geq 10^4 C_G(A + 1)(\kappa L)^2$; $Q_G^{6c} \leq x$ (as $\kappa L \geq 6c$); $\log x \geq 16$; and $x \geq C_4 A Z^4$, with C_4 as large as needed below.

Character expansion. As $\gcd(Q, D) = 1$, every character χ modulo QD is $\chi = \chi_Q \chi_D$ with χ_Q modulo Q and χ_D modulo D . The function $f(n) = B(n) \mathbf{1}[n \equiv 1 \pmod{Q}]$ on units modulo QD satisfies $f(n) = \sum_{\chi \bmod QD} c(\chi) \chi(n)$, where, summing over the class 1 mod Q ,

$$c(\chi) = \frac{1}{\varphi(QD)} \sum_{n \in (\mathbb{Z}/QD)^\times} f(n) \bar{\chi}(n) = \frac{\mathbb{E}_D[B \bar{\chi}_D]}{\varphi(Q)}.$$

Consequently:

- (a) $c(\chi) = 0$ unless the conductor of χ_D divides some d_i . Indeed, the mean of $\mathbf{1}[n \equiv b_i \pmod{d_i}] \bar{\chi}_D(n)$ over units mod D vanishes unless χ_D is trivial on $\{n \equiv 1 \pmod{d_i}\}$. So $c(\chi) \neq 0$ implies that the conductor of χ is at most $Qd_i \leq Z \leq Q_G$.
- (b) $|c(\chi)| \leq \mathbb{E}_D |B| / \varphi(Q) = A\mu / \varphi(Q)$, and $c(\chi_0) = \mu / \varphi(Q)$ for the principal character χ_0 .
- (c) If χ is real and χ_D is induced by a primitive ψ of conductor $f > 1$, then $c(\chi) = \mathbb{E}_D [B\psi] / \varphi(Q) = \mu_\psi / \varphi(Q)$. For the last equality: a cell with $f \nmid d_i$ has ψ -mean 0 by the argument in (a). If moreover $c(\chi) \neq 0$, then $f \mid d_i$ for some i by (a), $\gcd(f, Q) = 1$ as $f \mid D$, and the twist condition applies to ψ .

Let χ^* be the primitive character inducing χ . For a fixed modulus the map $\chi \mapsto \chi^*$ is injective, so by (a) at most $\sum_{q \leq Z} \varphi(q) \leq Z^2$ characters have $c(\chi) \neq 0$.

The prime sum. Let $S(x) = \sum_{p \leq x, p \nmid QD} f(p) \log p = \sum_{\chi} c(\chi) \vartheta(x; \chi)$. Since $|\vartheta(x; \chi) - \vartheta(x; \chi^*)| \leq \log(QD) \leq 2Z$ (as $D \mid \text{lcm}(1, \dots, \max d_i)$), so

$\log D \leq 1.04 \max d_i$),

$$S(x) = \frac{\mu x}{\varphi(Q)} + \sum_{\chi: c(\chi) \neq 0} c(\chi) (\vartheta(x; \chi^*) - E_0(\chi^*)x) + R_1, \quad (3)$$

where $|R_1| \leq Z^2 \cdot (A\mu/\varphi(Q)) \cdot 2Z = 2AZ^3\mu/\varphi(Q)$. Here $E_0(\chi^*) = 1$ only for $\chi = \chi_0$, and $|R_1| \leq \mu x/(400\varphi(Q))$ as $x \geq 800AZ^3$.

Case 0: no exceptional zero in the sense of Theorem 3.2. The χ^* in (3) are distinct primitive characters of conductor $\leq Q_G$. By (b) and Theorem 3.2, the middle sum is at most

$$\frac{A\mu}{\varphi(Q)} C_G x \left(e^{-L} + \frac{(\kappa L)^2}{Q_G} \right) \leq \frac{\mu x}{\varphi(Q)} \left(\frac{1}{400} + \frac{1}{10^4} \right) \leq \frac{\mu x}{200\varphi(Q)},$$

using $\log x / \log Q_G = \kappa L \geq L$. So $S(x) > 0$.

Exceptional case: a zero β_1 with character χ_1 and $1 - \beta_1 < 1/(\kappa \log Q_G)$. Put $u = (1 - \beta_1) \log x$; then $u < \log x / (\kappa \log Q_G) = L$, and $\beta_1 > 1/2$. The replaced right-hand side of Theorem 3.2 is $C_G u x (e^{-\kappa L} + \kappa L / Q_G)$. Since $u \leq \min(u, 1) L$ (recall $L \geq 2$), its product with $A\mu/\varphi(Q)$ is at most

$$\frac{\mu x}{\varphi(Q)} \min(u, 1) \cdot C_G A L \left(e^{-\kappa L} + \frac{\kappa L}{Q_G} \right) \leq \frac{\mu x}{200\varphi(Q)} \min(u, 1).$$

Here $C_G A L e^{-3L} \leq C_G A e^{-L} \leq 1/400$, and $C_G A \kappa L^2 / Q_G \leq 10^{-4}$. If no χ with $c(\chi) \neq 0$ has $\chi^* = \chi_1$, this bounds the middle sum of (3), and $S(x) > 0$ as in Case 0. Otherwise, by injectivity, exactly one such χ exists. It is real, because χ_1 is; hence $c(\chi)$ is real, as B and χ_D are. Its term is $c(\chi)(\vartheta(x; \chi_1) + x^{\beta_1}/\beta_1) - c(\chi)x^{\beta_1}/\beta_1$. The first part is covered by the bound just proved.

Case A: χ_D is principal. Then $c(\chi) = \mu/\varphi(Q)$, and the conductor q_1 of χ_1 divides Q , so $q_1 \leq Z$. The main terms combine to $\lambda \mu x / \varphi(Q)$ with $\lambda = 1 - x^{\beta_1-1}/\beta_1$. Since $1/\beta_1 \leq 1 + 2(1 - \beta_1)$, $1 - e^{-u} \geq (1 - e^{-1}) \min(u, 1)$ and $2(1 - \beta_1)e^{-u} = 2ue^{-u}/\log x \leq 2 \min(u, 1)/\log x$,

$$\lambda \geq 1 - e^{-u} - 2(1 - \beta_1)e^{-u} \geq (0.63 - 2/16) \min(u, 1) \geq \min(u, 1)/2.$$

So the error from Theorem 3.2 is at most $\lambda \mu x / (100\varphi(Q))$. By Theorem 3.1, $u \geq 16(1 - \beta_1) \geq 16/(c_2 Z^{1/2}(\log Z)^2)$, so $1/\lambda \leq 2 + c_2 Z^{1/2}(\log Z)^2/8 \leq C_5 Z$. Hence $|R_1| \leq 2AZ^3\mu/\varphi(Q) \leq \lambda \mu x / (100\varphi(Q))$ once $x \geq 200C_5 AZ^4$. Altogether $S(x) \geq (1 - 2/100)\lambda \mu x / \varphi(Q) > 0$.

Case B: χ_D is not principal. Then χ_D is real and is induced by a real primitive ψ of conductor $f > 1$. By (c) and the twist condition, $|c(\chi)x^{\beta_1}/\beta_1| \leq (\mu/(4\varphi(Q))) \cdot 2x = \mu x / (2\varphi(Q))$, and $S(x) \geq (1 - 1/2 - 1/200 - 1/400)\mu x / \varphi(Q) > 0$.

In all cases $S(x) > 0$. So some prime $p \leq x$ with $p \nmid QD$ and $p \equiv 1 \pmod{Q}$ has $B(p) > 0$, and then $W(p) > T$. $\square$

Remark 4.2. This is Linnik's theorem with weights. Every character coefficient is bounded by $\mathbb{E}_D |B| / \varphi(Q)$, not by the ℓ^1 mass of a cell representation. Theorem 3.2 controls all characters of conductor $\leq Q_G$ simultaneously, so

the number of moduli d_i never multiplies the error. In [2] and in the first version of this note, each progression was instead treated separately with the Thorner–Zaman theorem [30]. That costs $\log x \gg K \max(\log Z, K)$ with $K = 1 + \log(\sum |c_i|/\varphi(d_i)/\mu)$, which here is of order $(\log T)^{14}$ instead of $(\log T)^7$. No lower bound for $1 - \beta_1$ is needed in Case B. In Case A we use the effective bound of Theorem 3.1 only to absorb the term R_1 .

5. PRODUCT SYSTEMS AND THE SANDWICH MINORANT

Setting. Let $\mathcal{P}$ be a finite set of N coordinates. Let X_ℓ , $\ell \in \mathcal{P}$, be independent random variables, X_ℓ uniform on a finite set G_ℓ with $|G_\ell| \leq T$; write $\mathbb{P}, \mathbb{E}$ for this product measure. An *event* is a set $E = \{X_\ell = \sigma_\ell \mid \ell \in \text{supp } E\}$ with $\emptyset \neq \text{supp } E \subseteq \mathcal{P}$ and $\sigma \in \prod_{\ell \in \text{supp } E} G_\ell$. Let $E_1, \dots, E_m$ be events (repetitions allowed) with $|\text{supp } E_i| \leq k$, and put $A_i = \mathbf{1}_{E_i}$, $F_{<i} = \prod_{j < i} (1 - A_j)$, $F = F_{<m+1}$,

$$w_\ell = \sum_{i: \ell \in \text{supp } E_i} \mathbb{P}(E_i), \quad S = \sum_{i \leq m} \mathbb{P}(E_i).$$

By Lemma 2.7, the residual system of Definition 2.6 is an instance with $N \leq T$ and $m \leq T^{k+2}$.

Lemma 5.1 (Local lemma with conditioning; classical). *Let $\mathcal{A}$ be a finite family of events in a product space, each depending on a finite set of coordinates; write $A' \sim A$ if they share a coordinate. Suppose $x_A \in [0, 1)$ satisfy $\mathbb{P}(A) \leq x_A \prod_{A' \sim A, A' \neq A} (1 - x_{A'})$. Then $\mathbb{P}(\bigcap_{A \in \mathcal{A}} \bar{A}) \geq \prod_{A \in \mathcal{A}} (1 - x_A)$. Moreover, for every event B depending on a set J of coordinates, $\mathbb{P}(B \mid \bigcap_{A \in \mathcal{A}} \bar{A}) \leq \mathbb{P}(B) \prod_{A \in \Gamma(B)} (1 - x_A)^{-1}$, where $\Gamma(B)$ is the set of $A \in \mathcal{A}$ depending on some coordinate in J .*

Proof. The first statement is [1, Lemma 5.1.1]. Its proof shows $\mathbb{P}(A \mid \bigcap_{A' \in \mathcal{S}} \bar{A}') \leq x_A$ for every $\mathcal{S} \subseteq \mathcal{A}$ with $A \notin \mathcal{S}$. For the second (cf. [14, Thm. 2.1]), note that B is independent of $\bigcap_{A \notin \Gamma(B)} \bar{A}$, so

$$\mathbb{P}\left(B \mid \bigcap_{A \in \mathcal{A}} \bar{A}\right) \leq \frac{\mathbb{P}(B) \mathbb{P}(\bigcap_{A \notin \Gamma(B)} \bar{A})}{\mathbb{P}(\bigcap_{A \in \mathcal{A}} \bar{A})} = \frac{\mathbb{P}(B)}{\mathbb{P}(\bigcap_{A \in \Gamma(B)} \bar{A} \mid \bigcap_{A \notin \Gamma(B)} \bar{A})}.$$

The chain rule and the displayed claim bound the denominator below by $\prod_{A \in \Gamma(B)} (1 - x_A)$. $\square$

Corollary 5.2 (Proved). *In the Setting, assume $w_\ell \leq 1/(64k)$ for all ℓ . Then $\delta := \mathbb{E}F \geq e^{-2.2S}$. Moreover, for every subfamily $\mathcal{A}$ of $\{E_1, \dots, E_m\}$ and every event B on at most k coordinates, $\mathbb{P}(B \mid \text{no } A \in \mathcal{A} \text{ occurs}) \leq 1.04 \mathbb{P}(B)$.*

Proof. Take $x_{E_i} = 2\mathbb{P}(E_i) \leq 2w_\ell \leq 1/32$ for any $\ell \in \text{supp } E_i$. For an event B on a set J of coordinates, $\sum_{A \in \Gamma(B)} x_A \leq 2 \sum_{\ell \in J} w_\ell \leq |J|/(32k)$. With $B = E_i$ this gives $\prod_{A \sim E_i} (1 - x_A) \geq 31/32$, so the hypothesis of Lemma 5.1 holds for every subfamily. Since $1 - x \geq e^{-1.1x}$ for $0 \leq x \leq 1/32$, we get

$\delta \geq \prod_i (1 - 2\mathbb{P}(E_i)) \geq e^{-2.2S}$, and $\prod_{\Gamma(B)} (1 - x_A)^{-1} \leq e^{1.1/32} \leq 1.04$ when $|J| \leq k$. $\square$

Efron–Stein decomposition (classical; [8, 18], [25, §8.3]). For $\varphi : \prod_{\mathcal{P}} G_\ell \rightarrow \mathbb{R}$ and $W \subseteq \mathcal{P}$ let $\mathbb{E}[\varphi \mid X_W]$ be the conditional expectation given the coordinates in W , and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid X_W].$$

Then $\varphi = \sum_U \varphi^{=U}$, the components are mutually orthogonal, and $\varphi^{=U}$ is orthogonal to every function of $X_{U'}$ with $U' \not\supseteq U$. Hence the space of sums of functions of at most t coordinates is $\bigoplus_{|U| \leq t} V_U$, where V_U is the space of all possible $\varphi^{=U}$, and

$$\begin{aligned} \text{En}(\varphi; t) &:= \sum_{|U| > t} \|\varphi^{=U}\|_2^2 \\ &= \min \left\{ \mathbb{E}(\varphi - g)^2 : g \text{ a sum of functions of } \leq t \text{ coordinates} \right\}, \end{aligned} \tag{4}$$

attained at $\varphi^{\leq t} := \sum_{|U| \leq t} \varphi^{=U}$.

For $j \leq m$ let $F^{(j)}$ be the function of the coordinates outside $\text{supp } E_j$ obtained from $F_{<j}$ by fixing $X_\ell = \sigma_{j,\ell}$ for $\ell \in \text{supp } E_j$. Then $A_j F_{<j} = A_j F^{(j)}$, and $F^{(j)}$ is the avoidance indicator of the events E_i ($i < j$) compatible with E_j , restricted to $\mathcal{P} \setminus \text{supp } E_j$. Restricting such an event may leave it with empty support, and then $F^{(j)} \equiv 0$.

Lemma 5.3 (Bazzi–Razborov sandwich [3, 28]; proved). *Let $u_1, \dots, u_m$ be arbitrary real functions, $v_i = \sum_{j < i} A_j u_j$, and*

$$B = 1 - \sum_{i \leq m} A_i (1 - v_i)^2.$$

Then $B \leq F$ pointwise, and with $e_j = F_{<j} - u_j$,

$$F - B = \sum_i A_i \left(\sum_{j < i} A_j e_j \right)^2, \quad \mathbb{E}[F - B] \leq m^2 \sum_{j \leq m} \mathbb{E}[A_j e_j^2].$$

If u_j is a function of the coordinates outside $\text{supp } E_j$, then $\mathbb{E}[A_j e_j^2] = \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$.

Proof. The identity $1 - F = \sum_i A_i F_{<i}$ holds pointwise: the sum counts the first occurring event. Put $f_{<i} = 1 - F_{<i} = \sum_{j < i} A_j F_{<j}$. If $F_{<i} = 1$, then $A_j = 0$ for all $j < i$, so $v_i = 0$ and $A_i(1 - v_i)^2 = A_i = A_i F_{<i}$. If $F_{<i} = 0$, then $f_{<i} = 1$, and $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$. In both cases $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$, and summing over i ,

$$F - B = \sum_i A_i [(1 - v_i)^2 - F_{<i}] = \sum_i A_i (f_{<i} - v_i)^2 \geq 0,$$

with $f_{<i} - v_i = \sum_{j<i} A_j(F_{<j} - u_j)$. By Cauchy–Schwarz, $(\sum_{j<i} A_j e_j)^2 \leq m \sum_{j<i} A_j e_j^2$. Drop $A_i \leq 1$ and sum over i . For the last claim, $A_j e_j^2 = A_j(F^{(j)} - u_j)^2$, and $F^{(j)} - u_j$ is independent of A_j . $\square$

This is Razborov’s simplification [28] of Bazzi’s argument [3]. The choice of u_j as an ℓ^2 -approximation of the “conditional survival function” $F^{(j)}$ is the one used there. We include the short proof for completeness. From now on

$$u_j := (F^{(j)})^{\leq t}, \quad \text{so} \quad \mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \text{En}(F^{(j)}; t). \quad (5)$$

Cells. Now let $G_\ell = (\mathbb{Z}/\ell^{e_\ell})^\times$ with $\ell^{e_\ell} \leq T$, as in Definition 2.6. A *cell* on $I \subseteq \mathcal{P}$ is a set $C = \{X_I = c\}$, $c \in \prod_I G_\ell$. On integers n (with $X_\ell = n \bmod \ell^{e_\ell}$) it is the unit class $n \equiv c \pmod{d_C}$, $d_C = \prod_I \ell^{e_\ell} \leq T^{|I|}$, and $\mathbb{P}(C) = 1/\varphi(d_C)$. If C, C' are cells on I, J , then $\mathbf{1}_C \mathbf{1}_{C'} = \mathbf{1}_{C \cap C'}$, where $C \cap C'$ is empty or a cell on $I \cup J$, both as functions on $\prod G_\ell$ and on $\mathbb{Z}$. The Haar mean of a function of the coordinates is its mean over units modulo any D divisible by the moduli of its cells: $\mathbb{E} = \mathbb{E}_D$ in the notation of §4.

Lemma 5.4 (Cell structure of B ; proved). *With u_j as in (5), B is a finite real combination of cells, each on at most $3k + 2t$ coordinates. So each modulus satisfies $d \leq T^{3k+2t}$, it is composed of free primes, and the class is a unit class. The identity $B = 1 - \sum_i A_i(1 - v_i)^2$ also holds for the expanded cell combination evaluated at any integer n .*

Proof. For $\varphi = F^{(j)}$, a function of the coordinates outside $\text{supp } E_j$,

$$\varphi^{\leq t} = \sum_{|U| \leq t} \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid X_W],$$

and $\mathbb{E}[\varphi \mid X_W] = \sum_{C \text{ cell on } W} \mathbb{E}[\varphi \mid C] \mathbf{1}_C$. So u_j is a combination of cells on at most t coordinates. Expanding,

$$B = 1 - \sum_i A_i + 2 \sum_i \sum_{j<i} A_i A_j u_j - \sum_i \sum_{j, j' < i} A_i A_j A_{j'} u_j u_{j'}.$$

Each term is a product of at most three A ’s, each a single cell on $\leq k$ coordinates, and at most two u ’s. So every cell lies on at most $3k + 2t$ coordinates. The last claim holds because the cell identities hold on $\mathbb{Z}$. $\square$

Lemma 5.5 (ℓ^1 -tightness; proved). *If $B \leq F$ pointwise, $F \geq 0$ and $\mathbb{E}[F - B] \leq \eta \mathbb{E}B$, then $\mathbb{E}|B| \leq (1 + 2\eta) \mathbb{E}B$.*

Proof. $B^- := \max(-B, 0) \leq F - B$ since $F \geq 0$, and $\mathbb{E}|B| = \mathbb{E}B + 2\mathbb{E}B^-$. $\square$

Lemma 5.6 (Twist; proved). *In the situation of Definition 2.6, assume $w_\ell \leq 1/(64k)$ for all $\ell \in \mathcal{P}$ and $\mathbb{E}[F - B] \leq \delta/100$, where $\delta = \mathbb{E}F$. Write $B = \sum_i c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ as in Lemma 5.4, and let $\mu = \sum_i c_i/\varphi(d_i) = \mathbb{E}B$.*

Let ψ be a real primitive character of conductor $f > 1$ with $\gcd(f, Q_\Pi) = 1$ and $f \mid d_i$ for some i . Then $\mu \geq 0.99\delta$ and

$$|\mu_\psi| \leq \mu/4, \quad \mu_\psi = \sum_{i: f \mid d_i} c_i \psi(b_i) / \varphi(d_i).$$

Proof. $\mu = \mathbb{E}F - \mathbb{E}[F - B] \geq 0.99\delta$. As $2 \nmid f$ and ψ is real primitive, f is squarefree. Its prime factors are free, since they divide some d_i . View $\psi(n)$ as a function of the coordinates X_ℓ , $\ell \mid f$. For a cell C with modulus d , $\mathbb{E}[\mathbf{1}_C \psi] = \psi(b)/\varphi(d)$ if $f \mid d$. Otherwise some $\ell \mid f$ has X_ℓ free in C , and the nontrivial ℓ -component of ψ averages to 0. Hence $\mu_\psi = \mathbb{E}[B\psi]$, and

$$|\mu_\psi| \leq \mathbb{E}|B - F| + |\mathbb{E}[F\psi]| = \mathbb{E}[F - B] + |\mathbb{E}[F\psi]|.$$

Fix a prime $\ell_0 \mid f$ and let F' be the avoidance indicator of the events E_i with $\ell_0 \notin \text{supp } E_i$. Given $X_{-\ell_0} = (X_\ell)_{\ell \neq \ell_0}$, we have $F = F' \cdot \mathbf{1}[X_{\ell_0} \notin \text{Forb}]$, where Forb is the set of values σ_{i, ℓ_0} of the events $E_i \ni \ell_0$ whose other conditions hold. The ℓ_0 -component ψ_0 of ψ is a nontrivial character of X_{ℓ_0} , and $F' \cdot \prod_{\ell \neq \ell_0} \psi_\ell$ depends on $X_{-\ell_0}$ only. So

$$\begin{aligned} |\mathbb{E}[F\psi]| &= \left| \mathbb{E} \left[F' \prod_{\ell \neq \ell_0} \psi_\ell \mathbb{E}_{X_{\ell_0}} [\psi_0 \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \right] \right| \\ &\leq \sum_{i: \ell_0 \in \text{supp } E_i} \frac{\mathbb{P}(F' = 1, E_i \setminus \ell_0)}{|G_{\ell_0}|}, \end{aligned}$$

where $E_i \setminus \ell_0$ is the event given by the conditions of E_i off ℓ_0 . By Corollary 5.2 applied to the family defining F' , $\mathbb{P}(E_i \setminus \ell_0 \mid F' = 1) \leq 1.04 \mathbb{P}(E_i \setminus \ell_0)$. Hence $|\mathbb{E}[F\psi]| \leq 1.04 w_{\ell_0} \mathbb{E}F' \leq 0.02 \mathbb{E}F'$. Likewise $\mathbb{E}F' - \mathbb{E}F = \mathbb{E}[F' \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \leq 0.02 \mathbb{E}F'$, so $\mathbb{E}F' \leq \delta/0.98$. Altogether $|\mu_\psi| \leq 0.01\delta + 0.0205\delta \leq 0.031\mu/0.99 < \mu/4$. $\square$

6. THE EFRON-STEIN TAIL VIA THE SWITCHING LEMMA

For $f : \{0, 1\}^n \rightarrow \mathbb{R}$ write $f = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S$ with $\chi_S(x) = (-1)^{\sum_{s \in S} x_s}$, and $W^{\geq d}[f] = \sum_{|S| \geq d} \hat{f}(S)^2$. A p -random restriction $\rho = (J, z)$ chooses each coordinate independently to be free (in J) with probability p , and fixes the others to independent uniform bits z . The restricted function f_ρ is a function of x_J . $\text{DT}(g)$ is the decision-tree depth of g . A decision tree of depth $< s$ computes a function of Fourier degree $< s$.

Theorem 6.1 (Håstad's switching lemma [15]; cited in the form of [25, §4.4]). *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be computed by a DNF of width at most w , and let ρ be a p -random restriction. Then $\Pr_\rho[\text{DT}(f_\rho) \geq s] \leq (5pw)^s$ for every $s \geq 0$.*

Any absolute constant in place of 5 would do. We also use two classical facts. (1) For fixed J and $S \subseteq J$, the coefficient of $\widehat{f_{(J,z)}}$ at S is $\sum_{U \cap J = S} \hat{f}(U) \chi_{U \setminus J}(z)$. By Parseval in z , therefore, $\mathbb{E}_z[\widehat{f_{(J,z)}}(S)^2] = \sum_{U \cap J = S} \hat{f}(U)^2$. (2) A median of the binomial law $\text{Bin}(n, p)$ is $\lfloor np \rfloor$ or $\lceil np \rceil$ [19], so $\Pr[\text{Bin}(n, p) \geq \lfloor np \rfloor] \geq 1/2$.

Lemma 6.2 (Fourier tail; proved modulo Theorem 6.1). *In the Setting of §5, let φ be the avoidance indicator of a family of events, each on at most k coordinates, on a set of $N' \leq N$ coordinates. Let b be an integer with $2^b \geq 4NT$. Then for every integer $k_0 \geq 1$,*

$$\text{En}(\varphi; 10kbk_0) \leq 4 \cdot 2^{-k_0}.$$

Proof. If some event has empty support then $\varphi \equiv 0$, and there is nothing to prove. Otherwise:

(a) *Encoding.* For each coordinate ℓ enumerate $G_\ell = \{g_0, \dots, g_{q-1}\}$ ($q = q_\ell \leq T$) and put $\pi_\ell(u) = g_{\lfloor q \cdot \text{int}(u)/2^b \rfloor}$ for $u \in \{0, 1\}^b$, where $\text{int}(u) \in [0, 2^b)$ is the integer with binary digits u . Each fibre $\pi_\ell^{-1}(g_a)$ corresponds to an interval of integers. Since $2^b \geq q$, the fibre is nonempty, and it has $\lfloor 2^b/q \rfloor$ or $\lceil 2^b/q \rceil$ elements. An interval in $[0, 2^b)$ is a disjoint union of at most $2b$ dyadic intervals, i.e. of subcubes of $\{0, 1\}^b$ given by fixing at most b bits. Let $U = (U_\ell)$ be uniform on $(\{0, 1\}^b)^{N'}$, put $\pi(U) = (\pi_\ell(U_\ell))_\ell$ and $\tilde{\varphi} = \varphi \circ \pi$. The event $\{X_\ell = \sigma_\ell \ (\ell \in \text{supp } E)\}$ pulls back to a conjunction over $\ell \in \text{supp } E$ of disjunctions of conjunctions of at most b literals in the bits of U_ℓ . Distributing, it is a DNF of width $\leq kb$. Hence $f := 1 - \tilde{\varphi}$, the OR of these over all events, is a DNF of width at most $w := kb \geq 1$.

(b) *Concentration* (Linial–Mansour–Nisan [20]). Take $p = 1/(10w)$ and $d = k_0/p = 10wk_0$. By Theorem 6.1, $\Pr_\rho[\text{DT}(f_\rho) \geq k_0] \leq 2^{-k_0}$. Since $W^{\geq k_0}[f_\rho] \leq \mathbb{E}f_\rho^2 \leq 1$ always, and $W^{\geq k_0}[f_\rho] = 0$ when $\text{DT}(f_\rho) < k_0$, fact (1) gives

$$2^{-k_0} \geq \mathbb{E}_\rho W^{\geq k_0}[f_\rho] = \sum_U \hat{f}(U)^2 \Pr[|U \cap J| \geq k_0] \geq \frac{1}{2} \sum_{|U| \geq d} \hat{f}(U)^2.$$

In the last step, $|U \cap J| \sim \text{Bin}(|U|, p)$ and $\lfloor p|U| \rfloor \geq k_0$, so fact (2) applies. As $\hat{\varphi}(U) = -\hat{f}(U)$ for $U \neq \emptyset$, the function $\tilde{g} = \sum_{|S| < d} \hat{\varphi}(S) \chi_S$ satisfies $\mathbb{E}(\tilde{\varphi} - \tilde{g})^2 \leq 2^{1-k_0}$.

(c) *Back to residues.* Put $g(x) = \mathbb{E}[\tilde{g}(U) \mid \pi(U) = x]$. The blocks U_ℓ are independent and π acts blockwise. Hence $\mathbb{E}[\chi_S(U) \mid \pi(U) = x]$ is the product over the blocks ℓ met by S of $\mathbb{E}[\chi_{S \cap \text{block } \ell}(U_\ell) \mid \pi_\ell(U_\ell) = x_\ell]$, a function of fewer than d coordinates. So g is a sum of functions of at most $d = 10kbk_0$ coordinates. Let π_* be the law of $\pi(U)$. As $\tilde{\varphi}$ is a function of $\pi(U)$, Jensen gives $\mathbb{E}_{\pi_*}(\varphi - g)^2 = \mathbb{E}[\mathbb{E}[\tilde{\varphi} - \tilde{g} \mid \pi(U)]^2] \leq 2^{1-k_0}$. Finally, the density of the uniform law relative to π_* is $\prod_\ell 2^b / (q_\ell |\pi_\ell^{-1}(x_\ell)|) \leq \prod_\ell (1 - q_\ell 2^{-b})^{-1} \leq (1 - 1/(4N'))^{-N'} \leq 4/3$. So by (4), $\text{En}(\varphi; d) \leq \mathbb{E}(\varphi - g)^2 \leq \frac{4}{3} \cdot 2^{1-k_0} \leq 4 \cdot 2^{-k_0}$. $\square$

The width kb enters t only linearly. Nothing about the events is used beyond their support size; in particular there is no codegree hypothesis.

Corollary 6.3 (Proved modulo Theorem 6.1). *In the Setting of §5 with $N \leq T$ and $m \geq 1$, assume $w_\ell \leq 1/(64k)$ for all ℓ . Let*

$$b = \lceil \log_2(4T^2) \rceil, \quad t = 10 k b k_0, \\ k_0 = \lceil 3S \log_2 e + \log_2(400 m^2(S+1)) \rceil,$$

and define B by Lemma 5.3 with u_j as in (5). Then $\mathbb{E}[F - B] \leq \delta/100$.

Proof. Here $\log_2$ is the binary logarithm, and $2^b \geq 4T^2 \geq 4NT$. Apply Lemma 6.2 to each $F^{(j)}$: it is the avoidance indicator of events on at most k coordinates. This gives $\text{En}(F^{(j)}; t) \leq 4 \cdot 2^{-k_0} \leq e^{-3S}/(100m^2(S+1))$. By (5), $\mathbb{E}[F - B] \leq m^2 S e^{-3S}/(100m^2(S+1)) \leq e^{-3S}/100$, and $\delta \geq e^{-2.2S}$ by Corollary 5.2. $\square$

7. PROOF OF THE MAIN THEOREMS

Theorem 7.1 (Proved modulo Theorems 3.1, 3.2 and 6.1). *Let T be large, $7 \leq z \leq T$, $k = \lfloor \mathcal{L}/\log z \rfloor$, and let Π be the output of Lemma 2.5 with $c_0 = 1/(64k)$. Put*

$$k_0 = \lceil 3S^* \log_2 e + \log_2(400 T^{2k+4}(S^* + 1)) \rceil, \quad t = 10 k \lceil \log_2(4T^2) \rceil k_0.$$

Then there is a prime $p \equiv 1 \pmod{840}$ with $p > T$, $W(p) > T$ and $\log p \leq C_6 \log Z$, where C_6 is absolute and

$$\log Z \leq (\pi(z) + 64k^2 S^*)\mathcal{L} + 2(3k + 2t + 1)\mathcal{L} + 4.$$

Proof. Take the residual system of Definition 2.6 for this Π . By Lemma 2.7, $N \leq T$, $m \leq T^{k+2}$, $S \leq S^*$, every support has at most k primes, and $w_\ell \leq 1/(64k)$. If $m = 0$ put $B = 1$. Otherwise define B by Lemma 5.3 with u_j as in (5) and t as above. Since $m^2 \leq T^{2k+4}$ and $S \leq S^*$, Lemma 6.2 gives the bound of Corollary 6.3 at this k_0 , so $\mathbb{E}[F - B] \leq \delta/100$. Then Lemma 5.6 gives $\mu = \mathbb{E}B \geq 0.99\delta > 0$ and the twist condition for every real primitive ψ with conductor coprime to Q_Π dividing some d_i . As $\mathbb{E}[F - B] \leq \mu/99$, Lemma 5.5 gives $A = \mathbb{E}|B|/\mu \leq 1 + 2/99 < 1.03$. By Lemma 5.4, every modulus is $d_i \leq T^{3k+2t}$.

Pointwise inequality on integers. For every integer n the values $A_i(n) \in \{0, 1\}$ (with $X_\ell = n \pmod{\ell^{e_\ell}}$) satisfy the hypotheses of Lemma 5.3, and the cell expansion of B is valid on $\mathbb{Z}$ (Lemma 5.4). So $B(n) \leq F(n)$, and by Lemma 2.7(iv), $B(n) \leq \mathbf{1}[W(n) > T]$ for every $n \equiv 1 \pmod{Q_\Pi}$.

Transfer. Let $R = \max(T, \max_i d_i)$ and choose a prime $\ell_{\text{aux}} \in (R, 2R]$. Apply Theorem 4.1 with $Q = Q_\Pi \ell_{\text{aux}}$. Every d_i is composed of free primes, which are $\leq T$ and outside Π , so $\gcd(d_i, Q) = 1$; the b_i are units. The inequality $B \leq \mathbf{1}[W > T]$ holds on $n \equiv 1 \pmod{Q}$. The twist condition holds, as a conductor coprime to Q is coprime to Q_Π . Also $\mathbb{E}_D |B|/\mu = A \leq 1.03 \leq Z^{1/4}$, since $\mathbb{E}_D$ is the Haar mean. Here $\log Z = \log Q_\Pi + \log \ell_{\text{aux}} + \log \max d_i \leq \log Q_\Pi + \log 2 + 2(3k + 2t)\mathcal{L}$, and $\log Q_\Pi \leq (\pi(z) + 64k^2 S^*)\mathcal{L} + \log 24$ by Lemmas 2.7(i) and 2.5. Theorem 4.1 with $\log x = C_2(1 + \log 1.03) \log Z$ gives a prime p with $p \equiv 1 \pmod{Q}$, $W(p) > T$ and

$\log p \leq C_6 \log Z$. Since $\ell_{\text{aux}} \mid p-1$ we have $p > \ell_{\text{aux}} > T$, and $840 \mid Q_{\Pi}$ gives $p \equiv 1 \pmod{840}$. $\square$

Proof of Theorem 1.1. We take $z = \mathcal{L}^2$. Then $k \leq \mathcal{L}/(2 \log \mathcal{L})$ and $\pi(z) \ll \mathcal{L}^2/\log \mathcal{L}$. By Lemma 2.4(b), $S^* \ll \mathcal{L}^4 \log \mathcal{L}$. Then $k_0 \ll S^* + k\mathcal{L} \ll \mathcal{L}^4 \log \mathcal{L}$ and $t \ll k\mathcal{L}k_0 \ll \mathcal{L}^6$. Also $\log Q_{\Pi} \ll (\mathcal{L}^2/\log \mathcal{L} + k^2 S^*)\mathcal{L} \ll \mathcal{L}^7/\log \mathcal{L}$, so $\log Z \ll \mathcal{L}^7$. Theorem 7.1 gives a hard prime $p > T$ with $W(p) > T$ and $\log p \leq C\mathcal{L}^7$, i.e. $W(p) > T \geq \exp(c(\log p)^{1/7})$. Applying this to a sequence $T_1 < T_2 < \dots$ with T_{j+1} larger than the prime found for T_j gives infinitely many distinct primes. $\square$

Proof of Theorem 1.2. Take $z = \mathcal{L}^2$ again, and use Lemma 2.4(a) instead of (b). Now $k_0, t, \log Q_{\Pi}$ and $\log Z$ are all at most $\mathcal{L}^{O(1)}(S^* + 1)$, so

$$\log \log p \leq \log(S^* + 1) + O(\log \mathcal{L}) \leq (\log 2 + o(1)) \mathcal{L}/\log \mathcal{L}.$$

Since $x \mapsto x/\log x$ is increasing for $x \geq e$, we get $\mathcal{L}/\log \mathcal{L} \geq Y := (1/\log 2 - o(1)) \log \log p$, hence $\mathcal{L} \geq Y \log \mathcal{L} \geq Y \log Y$. That is,

$$\log W(p) > \mathcal{L} \geq (1/\log 2 - o(1)) \log \log p \cdot \log \log \log p.$$

Infinitely many primes arise as before. $\square$

Remark 7.2 (Where the exponent comes from; Assessment). The certified bound is $\log p \ll \log Z \ll \log Q_{\Pi} + t\mathcal{L}$. The quarantine term is $\log Q_{\Pi} \ll \mathcal{L}^7/\log \mathcal{L}$, which is the size of the Haar-side bound $\log(1/\delta^*(T)) \ll \mathcal{L}^7/\log \mathcal{L}$ proved, modulo [10, Prop. 1.4], in [2]. The junta term $t\mathcal{L} \ll \mathcal{L}^7$ is larger by a logarithm. It comes from the bit width $b \asymp \mathcal{L}$ in Lemma 6.2, multiplied by $k, k_0 \asymp S^*$ and the cost $\mathcal{L}$ per modulus prime. So the exponent 7 is bookkeeping, not a barrier. With $z = \mathcal{L}^2$, however, the quarantine alone forces $\log Q_{\Pi} \gg \pi(z)\mathcal{L} \asymp \mathcal{L}^3/\log \mathcal{L}$. We make no claim about the true order of $W(p)$.

8. SUMMARY OF STATUS

- Proved: Lemmas 2.1–2.7, 5.3, 5.4, 5.5, 5.6; Corollary 5.2 (with the classical Lemma 5.1); Lemma 2.4(a) (with the classical divisor bound).
- Proved modulo [10, Prop. 1.4]: Lemma 2.4(b).
- Proved modulo Håstad’s switching lemma (Theorem 6.1, a textbook result): Lemma 6.2, Corollary 6.3.
- Proved modulo Gallagher’s theorem (Theorems 3.2 and 3.1, quoted from the draft [22]): the linear transfer Theorem 4.1; and, together with Håstad’s switching lemma (Theorem 6.1), Theorem 7.1 and Theorem 1.2.
- Proved modulo Gallagher’s theorem, [10, Prop. 1.4] and Håstad’s switching lemma: Theorem 1.1.
- Assessment: Remarks 4.2 (comparison with the Thorner–Zaman route) and 7.2.

- Not claimed: anything about the Erdős–Straus conjecture; optimality of the exponent $1/7$; any numerical instance.
- The Thorner–Zaman theorem is no longer used; it is cited only for comparison.

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